

CONTENT

CONTENT.....	1
1. Order and Trade Flow.....	2
2. Order Book metrics.....	2
3. Price-based features.....	3
4. Liquidation-driven features.....	4
5. Open Interest and Funding rate.....	5
6. Cross-market features.....	6

1. Order and Trade Flow

Trade Flow Imbalance (TFI)	$TFI_t^\delta = B_{[t-\delta, t]} - S_{[t-\delta, t]}$	Difference between buy and sell trade volume over a fixed window δ. Hypothesis: High positive TFI $\rightarrow$ buy pressure $\rightarrow$ maker selling on ask suffers adverse selection. Filter when $TFI > \text{threshold}$	Cont, Kukanov, Stoikov (2014); Albers et al. (2021)
TFI with nonlinear transformation (sigmoid)	$\sigma(TFI)$	Smooths the impact, because extreme flow is often uninformed (panic); after transformation, very large TFIs have smaller impact.	Albers et al. (2021) (Fragmentation, Price Formation)
One-sided net volume	$\text{NetVolume} = \sum_{\text{trades}} \text{side} \cdot \text{size}$	Cumulative directional volume (buy minus sell) over fixed windows. Used to detect pressure.	Bieganowski & Ślepaczuk (2026)
Volume synchronized probability of informed trading (VPIN)	$VPIN = \frac{\sum_{\tau} V_{\tau}^S - V_{\tau}^B }{nV}$	Токсичность потока. Гипотеза: высокий VPIN = высокая вероятность информированной торговли $\rightarrow$ нужно фильтровать больше сделок.	Easley, Lopez de Prado, O'Hara (2010, 2012)
Informational richness (IR)	$IR = \log\left(\frac{\text{number of LOB changes}}{\text{number of best bid/ask changes} + \epsilon}\right)$	Characterizes how often best prices change relative to order book events. The calculation window should be wide enough to allow a significant number of events (1 minute).	Kolm et al. (2023)

2. Order Book metrics

Queue imbalance	$I = \frac{q^B - q^A}{q^B + q^A}$	Normalized difference of volumes at best bid and ask. Predicts the direction of the next mid-price move. Hypothesis: $I > \text{threshold}$ (e.g., 0.5) → strong buying pressure → maker selling on ask loses.	Gould & Bonart (2015); Cartea et al. (2015)
Microprice, its deviation and volatility	$m = \frac{p_a q_a + p_b q_b}{q_a + q_b}$ $\text{deviation} = \frac{\text{microprice} - \text{mid}}{\text{half spread}}$	Volume-weighted best prices. Hypothesis: Positive deviation (>0.5) → price highly likely to go up. Maker selling on ask loses.	Stoikov (2018); Albers et al. (2021)
Volume at best levels	$D = \frac{q_a + q_b}{2}$	Average volume at best levels..	Cont, Kukanov, Stoikov (2014)
Spread	$\text{spread} = \text{best ask price} - \text{best bid price}$	Hypothesis: Wide spread → liquidity crisis → maker should avoid providing liquidity on widening spread (transition), but may profit on extreme widening (post-shock reversion).	

3. Price-based features

Lagged return (normalized)	$r_{t-\delta}$	Hypothesis: Positive return over the last hour (2 hours / 4 hours) → high probability of subsequent negative return (mean reversion). Maker can profit by providing liquidity on the opposite side.	De Nicola (2021); Borgards & Czudaj (2020, 2021)
Realized volatility	$RV = \sqrt{\frac{1}{n-1} \sum_i (r_i - r_{average})^2}$	Standard deviation of returns over a window. Hypothesis: Moderate volatility (RV 2-5%) → toxic (adverse selection). Extreme volatility (>10%) → post-shock reversion → maker can profit from the reversal. Filter trades when RV is in the medium range, allow when very low or very high.	Yin et al. (2021)

4. Liquidation-driven features

Cumulative liquidation volume	$CumLiq = \sum liq\ volume\ за\ окно\ (5\ c,\ 30\ c,\ \dots)$	Total volume of forced closures over a window (e.g., 30 sec, 2 min). Hypothesis: Liquidation spike ($CumLiq > 3\sigma$ from mean) → price temporarily deviates from fair value → maker on the opposite side profits upon reversal.	Ali (2025)
Direction of liquidation pressure	$LiqImbalance = \frac{BuyLiq - SellLiq}{TotalLiq}$	Net liquidation pressure. Hypothesis: $LiqImbalance > 0.6 \rightarrow$ temporary overbought → maker sells on ask with high chance of reversal.	Boon Chuan (2026)

5. Open Interest and Funding rate

Open Interest (OI)	total number of open contracts	Rising OI with stable price → position accumulation; falling OI → deleveraging.	Badawi et al. (2024);
Long-short ratio	$\frac{Long\ OI}{Short\ OI}$	Hypothesis: Ratio > 2 → extreme long dominance → high risk of liquidation cascade → maker avoids providing on that side.	Badawi et al. (2024);
Funding rate (absolute and sign)	F%	Periodic payment between long and short positions in perpetuals. Hypothesis: Extreme funding (e.g., >0.1% per 8h or <-0.1%) → overheating → correction soon. Maker on the opposite side wins. Limitation: we have no funding rate data (but maybe could try to obtain it).	Badawi et al. (2024);
Change in funding rate	$\Delta F = F_t - F_{t-\Delta t}$	Rate of change. Rapid rise precedes cascades.	Badawi et al. (2024);
OI–funding interaction	$Z = \frac{OI - \mu_{OI}}{\sigma_{OI}} \cdot \frac{F - \mu_F}{\sigma_F}$	Joint extremeness (e.g., both >2σ) → high risk of cascade.	Badawi et al. (2024);

6. Cross-market features

Mean divergence (deviation from moving average of price difference)	$Div_t^{i,j,\Delta} = d(p^i, p^j) - rolling^{\Delta}(d(p^i, p^j))$	Centered price difference between two markets. In our case, can be adapted for liquidation prices across different exchanges.	Albers et al. (2021)
Cross-exchange lead-lag feature	Use features from the leader market (Binance) to predict on the laggard market (Bybit).	OFI, imbalance from the leading exchange. Idea: Information arrives first on the more liquid exchange, then spreads. Hypothesis for maker: Strong buy volume on Binance → within 100-500 ms rise on Bybit → maker on Bybit can buy before the rise. In our case, can be adapted for liquidations across different exchanges.	Albers et al. (2021, 2025)